1 Linear Optimization Problems

1.1 Introduction

1.1.1 Linear Function and Linear Inequality

Definition 1.1.1. *A function $f(x_1, x_2, \cdots, x_n)$ is a **linear function** iff it can be written in the following form*

$$f(x_1, x_2, \cdots, x_n) = c_1x_1 + c_2x_2 + \cdots + c_nx_n$$

where $c_1, c_2, \cdots, c_n$ are constants.

Definition 1.1.2. *For any linear function $f(x_1, x_2, \cdots, x_n)$ and any constant number b, the inequalities $f(x_1, x_2, \cdots, x_n) \geq b$ and $f(x_1, x_2, \cdots, x_n) \leq b$ are **linear inequalities**.*

Definition 1.1.3. *A function that is not linear is called **nonlinear function**.*

Example 1.1.1. *Assume we have three variables x_1, x_2 and x_3.. Classify the following as linear functions, linear inequality or nonlinear functions.*

1. $3x_1 + 4x_2 - 7x_3 = 29$

   This function is linear in x_1, x_2 and x_3 (No product(s) of two or more variables; variables occur to first power only). It has constant coefficients: $c_1 = 3$, $c_2 = 4$ and $c_3 = -7$. Hence, a **linear function**.

2. $-2x_1 + 5x_2 - x_3 \leq 4$

   This is a linear function with constant coefficients: $c_1 = -2$, $c_2 = 5$ and $c_3 = -1$. It is also a linear inequality with $b = 4$.

3. $2x_1x_2 + 2x_3 + 3x_3^2 \leq 3$

   This is not a linear function because we have product of x_1 and x_2. We also have a power of 2 for variable x_3. It is not a linear inequality either.

4. $e^a x_1 + \ln(b)x_2 \geq x_3 + c$ where a, b and c are constants

   All the coefficients in the function are constants. The function is also linear in x_1, x_2 and x_3. Hence, this is a linear inequality.

5. $(x_1 + 2x_2 + 3x_3)(x_1 + x_2 - x_3) \geq 8$

   The expansion contains a power of 2 for some variables. So, its not a linear function. It is not a linear inequality either. Hence, it is a **nonlinear function**.

1.1.2 Matrices and Vectors

Definition 1.1.4. *A **matrix** is a rectangular array of mathematical objects such as numbers, variables and symbols, in rows and columns.*

An $m \times n$ matrix A is represented as follows:

$$\mathbf{A} = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \tag{1}$$

- Matrices provide a method of organizing, storing, and working with mathematical information. They are useful in organizing and manipulating large amounts of data. Also, matrices provide a useful tool for working with models based on systems of linear equations

- To represent the elements of matrix $\mathbf{A}$ as variables, we use the notation (a_{ij}). For example

$$A = (a_{ij}) = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix} \tag{2}$$

- The first subscript in a_{ij} indicates the row; the second identifies the column. The nota- tion $\mathbf{A} = (a_{ij})$ represents a matrix by means of a typical element.

- The matrix $\mathbf{A}$ in (2) has three rows and two columns, and we say that $\mathbf{A}$ is 3×2 or that the size of $\mathbf{A}$ is 3×2.

  Definition 1.1.5. *A **vector** is a matrix with a single row or column.*

- Elements in a vector are often identified by a single subscript; for example

$$\mathbf{x} = \overrightarrow{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

- As a convention, we use lowercase boldface letters for column vectors and lowercase boldface letters followed by the prime symbol ($\prime$) for row vectors; for example

$$x' = (x_1, x_2, x_3) = (x_1\ x_2\ x_3)$$

- (Row vectors are regarded as **transposes** of column vectors

  Definition 1.1.6. *The **transpose** of matrix $\boldsymbol{A}$ is denoted by $\boldsymbol{A}'$ and is defined as*

$$A' = (a_{ij})' = (a_{ji})$$

For example, if

$$A = \begin{pmatrix} 6 & -2 \\ 4 & 7 \\ 1 & 3 \end{pmatrix} \qquad A' = \begin{pmatrix} 6 & 4 & 1 \\ -2 & 7 & 3 \end{pmatrix}$$

- If the transpose of a matrix **A** is the same as the original matrix, that is, if $A' = A$ or equivalently $(a_{ji}) = (a_{ij})$, then the matrix **A** is said to be **symmetric**. For example

$$B = \begin{pmatrix} 3 & 2 & 6 \\ 2 & 10 & -7 \\ 6 & -7 & 9 \end{pmatrix}$$

is symmetric. Clearly, all symmetric matrices are **square**.

Definition 1.1.7. *The diagonal of a $p \times p$ square matrix $\boldsymbol{A} = (a_{ij})$ consists of the elements $a_{11}, a_{22}, \cdots a_{pp}$. If a matrix contains zeros in all off-diagonal positions, it is said to be a* ***diagonal matrix****; for example, consider the matrix*

$$\boldsymbol{D} = \begin{pmatrix} 8 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 \end{pmatrix}$$

which can also be denoted as

$$\boldsymbol{D} = diag(8, -3, 0, 4)$$

- We also use the notation diag(**A**) to indicate a diagonal matrix with the same diagonal elements as **A**; for example

$$A = \begin{pmatrix} 3 & 2 & 6 \\ 2 & 10 & -7 \\ 6 & -7 & 9 \end{pmatrix} \qquad diag(A) = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 9 \end{pmatrix}$$

Definition 1.1.8. *A diagonal matrix with a 1 in each diagonal position is called an* ***identity matrix****, and is denoted by* $\boldsymbol{I}$*; for example*

$$I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Definition 1.1.9. *An* ***upper triangular matrix*** *is a square matrix with zeros below the diagonal; for example,*

$$T = \begin{pmatrix} 7 & 2 & 3 & -5 \\ 0 & 0 & -2 & 6 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 0 & 8 \end{pmatrix}$$

A ***lower triangular matrix*** *is defined similarly.*

- We denote a vector of zeros by $\mathbf{0}$ and a matrix of zeros by $\mathbf{O}$; for example

$$\mathbf{0} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \qquad \mathbf{O} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

1.1.3 Operations on Matrices

Sum of Two Matrices or Two Vectors

If two matrices or two vectors are the same size, they are said to be conformal for addition. Their sum is found by adding corresponding elements. Thus, if $\mathbf{A}$ is $n \times p$ and $\mathbf{B}$ is $n \times p$, then $\mathbf{C} = \mathbf{A} + \mathbf{B}$ is also $n \times p$ and is found as $\mathbf{C} = (c_{ij}) = (a_{ij} + b_{ij})$; for example

$$\begin{pmatrix} 7 & -3 & 4 \\ 2 & 8 & -5 \end{pmatrix} + \begin{pmatrix} 11 & 5 & -6 \\ 3 & 4 & 2 \end{pmatrix} = \begin{pmatrix} 18 & 2 & -2 \\ 5 & 12 & -3 \end{pmatrix}$$

The difference $\mathbf{D} = \mathbf{A} - \mathbf{B}$ between two conformal matrices $\mathbf{A}$ and $\mathbf{B}$ is defined similarly: $\mathbf{D} = (d_{ij}) = (a_{ij} - b_{ij})$.
Two properties of matrix addition are given in the following theorem.

Theorem 1.1.1. *If $\boldsymbol{A}$ and $\boldsymbol{B}$ are both $n \times m$, then*

(i) $\boldsymbol{A} + \boldsymbol{B} = \boldsymbol{B} + \boldsymbol{A}$

(ii) $(\boldsymbol{A} + \boldsymbol{B})' = \boldsymbol{A}' + \boldsymbol{B}'$

Product of a Scalar and a Matrix
Any scalar can be multiplied by any matrix. The product of a scalar and a matrix is defined as the product of each element of the matrix and the scalar:

$$c\mathbf{A} = (ca_{ij}) = \begin{pmatrix} ca_{11} & ca_{12} & \cdots & ca_{1m} \\ ca_{21} & ca_{22} & \cdots & ca_{2m} \\ \vdots & \vdots & & \vdots \\ ca_{n1} & ca_{n2} & \cdots & ca_{nm} \end{pmatrix} \tag{3}$$

Since $ca_{ij} = a_{ij}c$, the product of a scalar and a matrix is commutative:

$$c\mathbf{A} = \mathbf{A}c$$

Product of Two Matrices or Two Vectors

In order for the product **AB** to be defined, the number of columns in **A** must equal the number of rows in **B**, in which case **A** and **B** are said to be **conformal for multiplication**. Then the $(ij)th$ element of the product $C = AB$ is defined as

$$c_{ij} = \sum_k a_{ik} b_{kj}$$

which is the sum of products of the elements in the ith row of **A** and the elements in the jth column of **B**. Thus we multiply every row of **A** by every column of **B**. If **A** is $n \times m$ and **B** is $m \times p$, then **C=AB** is $n \times p$. We illustrate matrix multiplication in the following example.

Example 1.1.2. *Let*

$$\boldsymbol{A} = \begin{pmatrix} 2 & 1 & 3 \\ 4 & 6 & 5 \end{pmatrix} \qquad \textit{and} \qquad \boldsymbol{B} = \begin{pmatrix} 1 & 4 \\ 2 & 6 \\ 3 & 8 \end{pmatrix}$$

Then

$$\boldsymbol{AB} = \begin{pmatrix} 2\cdot 1 + 1\cdot 2 + 3\cdot 3 & 2\cdot 4 + 1\cdot 6 + 3\cdot 8 \\ 4\cdot 1 + 6\cdot 2 + 5\cdot 3 & 4\cdot 4 + 6\cdot 6 + 5\cdot 8 \end{pmatrix} = \begin{pmatrix} 13 & 38 \\ 31 & 92 \end{pmatrix} \tag{4}$$

Also,

$$\boldsymbol{BA} = \begin{pmatrix} 18 & 25 & 23 \\ 28 & 38 & 36 \\ 38 & 51 & 49 \end{pmatrix}$$

If **A** is $n \times m$ and **B** is $m \times p$, where $n \neq p$, then **AB** is defined, but **BA** is not defined. If **A** is $n \times p$ and **B** is $p \times n$, then **AB** is $n \times n$ and **BA** is $p \times p$. In this case, of course, $\mathbf{AB} \neq \mathbf{BA}$. If **A** and **B** are both $n \times n$, then **AB** and **BA** are the same size, but, in general

$$\mathbf{AB} \neq \mathbf{BA}$$

1.2 Further Reading on Matrix Algebra

You are expected to read the article on Matrix Algebra (Uploaded on the course webpage on Google ClassRoom). Focus on **matrix determinant, Inverse, Trace and rank and their properties**. Upon reading the text, answer the following questions.

Exercise 1.

1. Let $\mathbf{A} = \begin{pmatrix} 7 & -3 & 2 \\ 4 & 9 & 5 \end{pmatrix}$

    (a) Find $\mathbf{A}'$

(b) Verify that $(\mathbf{A}')' = \mathbf{A}$

2. Let $\mathbf{A} = \begin{pmatrix} 2 & 4 \\ -1 & 3 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 3 \\ 2 & -1 \end{pmatrix}$

(a) Find $(\mathbf{A} - \mathbf{B})'$ and $(\mathbf{AB})'$.

(b) Find $tr(\mathbf{AB})$ and compare with $tr(\mathbf{BA})$

(c) Find $|\mathbf{AB}|$ and $(\mathbf{A}^{-1})^{-1}$

3. Let $\mathbf{A} = \begin{pmatrix} 3 & 2 & 1 \\ 6 & 4 & 2 \\ 12 & 8 & 4 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \\ -1 & 1 & -2 \end{pmatrix}$

(a) Show that $\mathbf{AB} = \mathbf{O}$.

(b) Find a vector $\mathbf{x}$ such that $\mathbf{Ax} = \mathbf{0}$

(b) Find $\mathbf{BAB}$

4. Let $\mathbf{A} = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \end{pmatrix}$; $\mathbf{B} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}$; $\mathbf{C} = \begin{pmatrix} 2 & 1 & 1 \\ 5 & -6 & -4 \end{pmatrix}$

(a) Is $\mathbf{AB} = \mathbf{CB}$?

(b) What are the ranks of $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$?